



TECHNICAL DATA SHEET

ProLine ABS-GF10 Filament

Product Overview

ProLine ABS-GF10 is a 3D printing filament based on an ABS material containing 10% glass fiber. It offers good dimensional stability, high rigidity and good high-temperature resistance, making it suitable for functional parts that need to stay stable under load and heat.

Key Features

- Dimensionally stable
- High rigidity
- Weather resistant

Material Properties

Physical Properties			
Property	Test Method	Unit	Value
Density	ISO 1183	g/cm ³	1.08-1.09
MFR (220°C / 10 kg)	ISO 1133	g/10 min	20-30
Moisture Absorption (23°C / 24 h)	ISO 62	%	< 1
Mechanical Properties			
Tensile strength (X-Y)	ISO 527	MPa	25.2-28.5
Elongation at break (X-Y)	ISO 527	%	3-3.5
Flexural strength (X-Y)	ISO 178	MPa	81-83.5
Flexural strength (Z)	ISO 178	MPa	45-48.6
Flexural modulus (X-Y)	ISO 178	MPa	3360-3456
Flexural modulus (Z)	ISO 178	MPa	1805-2075
Notched impact strength (Z)	ISO 180	kJ/m ²	6-7.3
Notched impact strength (X-Y)	ISO 180	kJ/m ²	8.2-8.6
Thermodynamic Properties			
HDT @ 0.455 MPa (66 psi)	ISO 75	°C	88
Continuous Use Temperature	IEC 60216	°C	85

Recommended Printing Parameters

Parameter	Setting
Print Temperature	250-270°C
Bed Temperature	90-110°C
Bed Materials	Tempered glass, BuildTak, Carbon fiber board
Nozzle Diameter	Ø 0.4 / 0.6 mm
Nozzle and gear material	High strength steel
Model Cooling Fan	0-20%
Printing Environmental Temperature	Room temperature to ~60°C

Drying & Handling Notes

- Pre-dry: 80°C, minimum 5 h (hot-air oven) to ensure print success rate and model quality
- Post-dry: 80°C, 1-3 h to improve part strength

Disclaimer

Product information and test results in this document were obtained under controlled laboratory conditions. Real-world performance may vary depending on processing, environment, and application. The customer is responsible for determining product suitability for the intended use and for ensuring compliance with all applicable laws and regulations. The Company accepts no liability for the use of this information and provides no warranties, including any implied warranties of merchantability or fitness for a particular purpose, to the extent permitted by law.

Test Specimens

Test sample dimensions and printing direction

Figure 1. Tensile testing specimen; ASTM D638 (ISO 527, GB/T 1040)

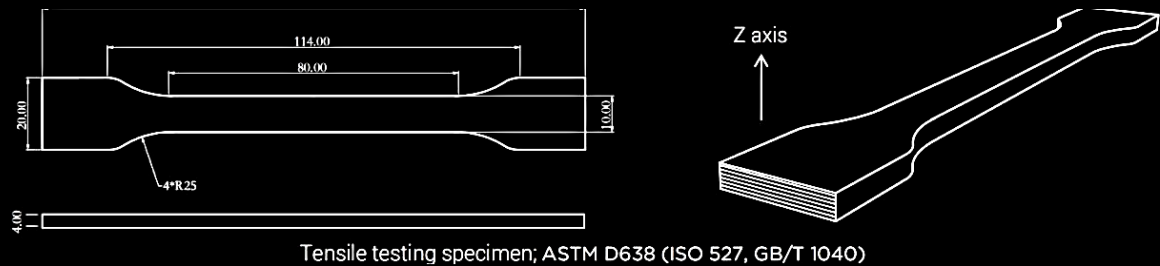


Figure 2. Flexural and impact testing specimens; ASTM D790 (ISO 178, GB/T 9341) and ASTM D256 (ISO 179, GB/T 1043)

